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LIGHT THERAPY SYSTEM

Abstract

A light therapy system is made up of a pressure cuff system having a cuff that is positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff and a light emitting system having a supporting structure and a light emitter positioned on the supporting structure, the light emitter being controllably powerable. The supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part. The light therapy system can further include a controller capable of controllably powering the light emitter, wherein controller can adjust the intensity or duration of the light directed onto the body part in response to an input or measurement related to a condition of the body part. A photometer can be provided separately or integrally to measure the condition of the body part. A method of providing light therapy is also provided.

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Background/Summary

PRIORITY [0001] The present application claims the benefit of domestic priority based on U.S. Provisional Patent Application 62/802,686 filed on Feb. 7, 2019, the entirety of which is incorporated herein by reference.

BACKGROUND

[0002] Photobiomodulation (PBM) or Low-level light therapy (LLLT) involves the application of laser and light-emitting diode (LED) light to living tissue to create a photobiostimulation effect. The application of laser and LED light to living tissue is performed using wavelengths between 600-1000 nm and power from 5-500 mW. Research has shown that the PBM accelerates wound healing, improves microcirculation, and improves muscle performance. In most studies, an infrared (IR) light source is placed on or near the surface of the body to produce a therapeutic dose of light at 0.5 to 5.0 J/cm². Various light sources include commercially available LEDs and lasers diodes packaged in integrated circuits. Some laser-based devices are pulsed to deliver higher instantaneous radiation intensity without thermally damaging the treated tissue.

[0003] A biphasic dose response effect is well established in the PBM treatment, with which, any therapeutic benefits are diminished above and below some optimal dosage range. Thus, accurate PBM dosing of muscles is critical for achieving benefits but confounded by limited and variable penetration depth of the IR light that is applied on the skin surface for treating the tissues beneath the skin surface.

[0004] Inside the tissue, photons are either scattered or absorbed. The therapeutic benefit is achieved when light is absorbed by mitochondria in the muscle. On most parts of the body, a portion of the IR light passes through the skin, and the underlying fat layer scatters the photons in a random pattern. Some photons continue to the underlying muscle fibers where the IR energy is absorbed by chromophores. The beneficial effects of the IR radiation of the muscle tissue increases mitochondrial membrane potential, oxygen consumption, and adenosine triphosphate (ATP) production. Since the IR light passing through the fat is randomly scattered, delivering a precise dose of the IR fluence to the body muscles is challenging given humans have varying levels of skin melanin and fat layer thickness throughout their bodies.

[0005] Near IR photometry is used in some commercial products (such as Lipid meter, SAT meter, FUTREX) to non-invasively measure skin fat thickness. The method uses light that is transmitted into the skin by IR emitting diodes. This radiation penetrates the tissue and is reflected, absorbed, and scattered according to the tissue's optical properties. The radiation emerging from the skin is detected at multiple distances between the source and detector of the optical measuring system and then processed to infer the amount of light absorbed by the skin and reflected back from the fat layer.

[0006] Intermittent pneumatic compression (IPC) devices are used medically to help prevent blood clots in the deep veins of the legs (deep vein thrombosis). These devices use pneumatic cuffs that

are usually placed around the legs. The cuffs fill with air and squeeze the legs. Upon release of the air, blood flows back through the veins and helps prevent blood clots. Recently, these devices have been used by athletes to improve circulation and facilitate muscle recovery after exercise. Also, the IPC devices are commercially available and frequently marketed to provide automated massage treatment. The IPC devices come in a variety of forms to treat different parts of the body and muscle groups. Leg units cover the foot and leg to the upper thigh. Shorts begin above the knee and extend upward above the waist. Some cuffs cover hands and arm up to the shoulder.

[0007] Therefore, there is a need for an improved light therapy system. There is a further need for a light therapy system with improved and/or more consistent dosing. There is a further need for a light therapy system with improved delivery of light to the muscles. There is a further need for improved and/or coordinated determination of fat layer thickness. There is a further need for a light therapy system that is used in conjunction with an intermittent pneumatic compression device.

SUMMARY

[0008] The present invention satisfies these needs. In one aspect of the invention, an improved light therapy system and improved method of providing therapeutic light is provided.

[0009] In another aspect of the invention, a light therapy system and method of use provides improved dosing and/or more consistent dosing.

[0010] In another aspect of the invention, a light therapy system and method of use provides improved delivery of light therapy to the muscles of a user.

[0011] In another aspect of the invention, a light therapy system and method of use utilizes fat layer thickness information for improved light therapy.

[0012] In another aspect of the invention, a light therapy system and method of use utilizes skin color information for improved light therapy.

[0013] In another aspect of the invention, an improved fat layer thickness determination and method is provided.

[0014] In another aspect of the invention, a light therapy system and method of use is used in conjunction with an intermittent pneumatic compression device.

[0015] In another aspect of the invention, a light therapy system comprises a pressure cuff system comprising a cuff that is positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff; and a light emitting system comprising a supporting structure and a light emitter positioned on the supporting structure, the light emitter being controllably powerable; wherein the supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part, and wherein the light emitter has a light emitting surface, the light emitting surface being a sufficient height from the supporting structure that the light emitting surface can be pressed into the skin and into a fatty layer of the body part when the cuff is pressurized.

[0016] In another aspect of the invention, a light therapy system comprises a pressure cuff system comprising a cuff that is positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff; and a light emitting system comprising a supporting structure and a light emitter positioned on the supporting structure, the light emitter being controllably powerable; wherein the supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part, and wherein the light emitter has a light emitting surface, the light emitting surface being a sufficient height from the supporting structure that the light emitting surface can be pressed into the skin and into a fatty layer of the body part when the cuff is pressurized, wherein the light emitter comprises a plurality of arrays of light emitters, each associated with a different muscle group of the body part.

[0017] In another aspect of the invention, a light therapy system comprises a pressure cuff system comprising a cuff that is positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff; and a light emitting system comprising a supporting structure and a light emitter positioned on the supporting structure, the

light emitter being controllably powerable; wherein the supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part, and wherein the light emitter has a light emitting surface, the light emitting surface being a sufficient height from the supporting structure that the light emitting surface can be pressed into the skin and into a fatty layer of the body part when the cuff is pressurized and further comprising a photometer capable of measuring a property of the body part.

[0018] In another aspect of the invention, a light therapy system comprises a pressure cuff system comprising a cuff that can be positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff; a light emitting member comprising a supporting structure and a light emitter positioned on the supporting structure, wherein the supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part; and a controller capable of controllably powering the light emitter, wherein the controller can adjust the intensity or duration of the light directed onto the body part in response to an input or measurement related to a condition of the body part.

[0019] In another aspect of the invention, a light therapy system comprises a pressure cuff system comprising a cuff that can be positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff; a light emitting member comprising a supporting structure and a light emitter positioned on the supporting structure, wherein the supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part; and a controller capable of controllably powering the light emitter, wherein the controller can adjust the intensity or duration of the light directed onto the body part in response to an input or measurement related to a condition of the body part, wherein the controller adjusts the intensity or duration of the light in response to a thickness of the fat layer of the body part.

[0020] In another aspect of the invention, a light therapy system comprises a pressure cuff system comprising a cuff that can be positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff; a light emitting member comprising a supporting structure and a light emitter positioned on the supporting structure, wherein the supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part; and a controller capable of controllably powering the light emitter, wherein the controller can adjust the intensity or duration of the light directed onto the body part in response to an input or measurement related to a condition of the body part and further comprising a photometer and wherein the controller adjusts the intensity or duration of the light in response to a signal from the photometer.

[0021] In another aspect of the invention, a light therapy system comprises a pressure cuff system comprising a cuff that can be positionable on or near a body part of a user and that can receive pressurized air to selectively pressurize and depressurize the cuff; a light emitting member comprising a supporting structure and a light emitter positioned on the supporting structure, wherein the supporting structure is positionable on an interior surface of the cuff so that the light emitter can direct light onto the body part; and a controller capable of controllably powering the light emitter, wherein the controller can adjust the intensity or duration of the light directed onto the body part in response to an input or measurement related to a condition of the body part, further comprising a plurality of photometers and wherein the controller adjusts the intensity or duration of the light in response to a signal from the photometers, wherein the photometer generates a signal in relation to the thickness of the fat layer of the body part or the transmissivity of the body part, wherein the light emitter comprises a plurality of arrays of light emitters, each associated with a different muscle group of the body part, and wherein each photometer is positionable on the interior surface of the cuff in proximity to a respective array.

[0022] In another aspect of the invention, a method of providing light therapy comprises providing a pressure cuff having a light emitter on an interior surface thereof; positioning the cuff on or near a

body part; inflating the cuff and pressing the light emitter directly or indirectly against the skin of the body part; determining the intensity or duration of light to be applied from the light emitter to the body part in relation to a condition of the body part; and powering the light emitter so that the light is applied to the body part at the determined intensity or duration.

[0023] In another aspect of the invention, a method of providing light therapy comprises providing a pressure cuff having a light emitter on an interior surface thereof; positioning the cuff on or near a body part; inflating the cuff and pressing the light emitter directly or indirectly against the skin of the body part; determining the intensity or duration of light to be applied from the light emitter to the body part in relation to a condition of the body part; and powering the light emitter so that the light is applied to the body part at the determined intensity or duration, further using a photometer to measure the condition of the body part and using the measurement to determine the intensity or duration of the light.

[0024] In another aspect of the invention, a light therapy system comprises a photometer having concentric rings of photometer light emitters and a central photodetector.

[0025] In another aspect of the invention, a method of providing light therapy comprises providing a photometer; using the photometer to determine a condition of a body part; and applying light therapy to the body part at an intensity or duration related to the condition of the body part.

Description

DRAWINGS

[0026] These features, aspects, and advantages of the present invention will become better understood with regard to the following description, appended claims, and accompanying drawings which illustrate exemplary features of the invention. However, it is to be understood that each of the features can be used in the invention in general, not merely in the context of the particular drawings, and the invention includes any combination of these features, where:

[0027] FIG. 1A is a schematic side view of a light therapy system according to the invention;

[0028] FIG. 1B is a schematic sectional view of the light therapy system of FIG. 1A in an unpressurized condition;

[0029] FIG. 1C is a schematic sectional view of the light therapy system of FIG. 1A in a pressurized condition;

[0030] FIG. 2A is a schematic top view of a version of a light emitter array of the light therapy system;

[0031] FIG. 2B is a schematic top view of another version of a light emitter array of the light therapy system;

[0032] FIG. 3A is a schematic top view of a monitoring system of the light therapy system;

[0033] FIG. 3B is a schematic side view of the monitoring system of FIG. 3A;

[0034] FIG. 3C is a schematic top view of another version of a monitoring system of the light therapy system;

[0035] FIG. 4 is a schematic side view of another version of a light therapy system of the invention;

[0036] FIG. 5 is a schematic open view of a version of an arrangement of light emitters of the light therapy system;

[0037] FIG. 6 is a schematic exploded view of a version of a pressure cuff system of the light therapy system;

[0038] FIG. 7 is a schematic representation of a display of a controller of the light therapy system;

[0039] FIG. 8A is a schematic exploded view of another version of a pressure cuff system of the light therapy system;

[0040] FIG. 8B is a schematic exploded view of another version of a pressure cuff system of the

light therapy system;

[0041] FIG. 9A is a chart showing fate of infrared photons for multiple simulations changing both melanin content ranging from fair to dark skin color and fat thickness from 3 mm to 10 mm; and [0042] FIG. 9B is a graph showing a profile of how light is re-emitted from the skin surface as a function of distance from a light emitter light source.

DESCRIPTION

[0043] The present invention relates to a light therapy system. In particular, the invention relates to a light therapy system including or utilizable with a pressure cuff. Although the invention is illustrated and described in the context of being useful in association with intermittent pneumatic compression, the present invention can be used in other ways, as would be readily apparent to those of ordinary skill in the art. Accordingly, the present invention should not be limited just to the examples and embodiments described herein.

[0044] FIG. 1A shows a light therapy system **100** of the present invention. The light therapy system **100** includes or can be useable with a pressure cuff system **105**. The pressure cuff system **105** includes a pressurizable cuff **110** that can be received around, adjacent, or in proximity to at least a portion of a part of a body, such as an extremity. In FIG. 1A, the cuff **110** is shown encircling a leg **115** of a user. The pressure cuff system **105** also includes a control system **120** that controls and/or monitors the pressurization of the cuff **110**. The control system **120** includes a pressure controller **125** that is capable of controlling the operation of an air pump **130**, such as an electrical air pump or a source of pressurize air, so that pressurized air may be passed through a line **135** leading to the cuff **110** to inflate or pressurize the cuff **110**. The pressure controller **130** can receive information from a pressure sensor **140** and can control the application of pressure to the cuff **110** in response to the received pressure data. The pressure controller **130** may also be capable of controlling a pressure gate **145** within the line **135**. The pressure gate **145** can include solenoids and/or venting mechanisms to allow the cuff **110** to be selectively pressurized and depressurized.

[0045] The light therapy system **100** also includes a light emitting system **150** positionable within the cuff **110** of the pressure cuff system **105** so that light can be administered to the portion of the leg **115** or other body part within the cuff **110**. The light emitting system **150** includes one or more light emitting members **155** that include a supporting structure **160** that supports one or more light emitters **165**. The light emitting system **150** is controllably connected, such as by electrical wire **170** and/or wireless technology, to a light emitting controller **175** of the control system **120**. The light emitting controller **175** is capable of controlling and/or monitoring the application of light, including the dose or dosage of light applied to the leg **115** or other body part. By dose or dosage of light applied it is meant the intensity and/or duration of the light application. Optionally, the light emitting controller **175** can be incorporated into or in communication with the pressure controller **125** so that the application of pressure and the application of light can be responsive to one another and/or coordinated together. The supporting structure **160** can take the form of a printed circuit board or the like through which each light emitter **165** can be powered and/or controlled.

[0046] FIGS. 1B and 1C show a sectional view of the light emitting system **150** positioned within the cuff **110** of the pressure cuff system **105** so that a light emitter **165** can be in proximity to the leg **115** or other body part and can deliver light to the body part. For example, the light emitting member **155** or the light emitters **165** can be positioned on the inside or on an interior surface of the cuff **110**. The cuff **110** is made of a flexible material and includes a hollow space **180** that is in communication with the line **135** of pressurized air. The hollow space **180** may be selectively pressurized and depressurized to inflate and deflate the cuff **110**. FIG. 1B shows the cuff in an unpressurized condition, and FIG. 1C shows the cuff in a pressurized condition. As can be seen, in the unpressurized condition, the light emitter **165** rests near or against the skin **185** of the leg **115** or other body part without significant deformation of the skin **185**. However, when in the pressurized condition shown in FIG. 1C, the pressure of the cuff **110** forces the light emitter **165** into the skin **185**. When sufficient pressure is applied, the light emitter **165** can be further pressed into the fatty

layer **186** so that it is in proximity to the muscle **187**. Optionally, a spacer **190** or the like can be provided if needed to increase the thickness of the light emitter **165** and to extend the distance of the light emitter from the supporting structure **160**. Also optionally, a cover layer **195** made of light transmitting material can be provided to cover the light emitter **165** so that it does not directly contact the skin **185**. Alternatively, the cover layer **195** can be removed and the light emitter can directly contact the skin **185**.

[0047] In one version, the light therapy system **100** may be used to administer therapeutic light to the leg **115** or other body part that is within or near the cuff **115**. For example, the light therapy system **100** may be used to administer photobiomodulation therapy, also known as low-level light therapy, where the application of laser or light emitting diode (LED) light to living tissue.

Therapeutic light that is administered to create a photobiostimulation effect when administered using wavelength of light between about 600 and 1000 nm at a power ranging from about 5 to 500 mW from each light source. The photobiostimulation effect has been shown to accelerate wound healing, improve circulation, and/or improve muscle performance. The light therapy system **100** of the present invention improves on previous attempts at light therapy by its inclusion within the pressure cuff system **105** which applies pressure to the light emitting system **150** to press the light emitter **165** into the skin **185** and closer to the muscle **187** or other target of therapy. By positioning the light emitter **165** closer to the muscle, a greater therapeutic effect can be achieved and/or greater efficiency is light application can be achieved. When light is applied to the surface of the skin **185**, photons are either scattered or absorbed. The scattering is especially prevalent in the fatty tissue **186**. The positioning of the light emitter **165** closer to the muscle **187** and the compression of the fatty layer **186** allows for increased absorption of the photons by the muscle and decreased scattering. The therapeutic effect is achieved when photons are absorbed by mitochondria in the muscle. The beneficial effect of the photon absorption is believed to include an increase in mitochondrial membrane potential, oxygen consumption, and adenosine triphosphate production.

[0048] In one version of the light therapy system, the one or more light emitters **165** comprises one or more light emitting diodes. In one particular version, the light emitters **165** are configured to emit light consisting of wavelengths in the near infrared range or from about 780 nm to about 1000 nm. While it is possible to use light outside this range, studies have shown that light in this range of wavelengths is preferred for muscle therapy since shorter wavelengths do not penetrate as far into tissue and longer wavelengths are not as effective at stimulating the muscle mitochondria.

[0049] In one version of the light therapy system **100**, the light emitters **165** are sized and shaped to penetrate into the fatty layer **186** when the cuff **110** is pressurized so that the light emitter can be in proximity to the underlying muscle **187**. The light emitters **165** have a light emitting surface that is in contact with the skin **185** either directly or through the cover layer **195**. In one version, the height of the light emitting surface above, i.e. away from, the support structure or an interior surface of the cuff **110** is at least about 1 mm, or at least about 2 mm, or at least about 3 mm, or at least about 5 mm, or at least about 10 mm. A spacer **190** can be provided as necessary or the light emitter **165** can be designed with the desired height. In one version, different height light emitters can be provided to accommodate different fat layer thickness of a user.

[0050] FIGS. 2A and 2B illustrate versions of light emitting members **155** that are designed to provide an array **200** of light emitters **165**. The array **200** can comprise a plurality of light emitters **165** arranged in any suitable pattern. By array of light emitters it is meant a group of a plurality of light emitters **165** that are powerable with by the light emitter controller at the same or similar intensity and duration. The array **200** can be on one printed circuit board or can be on multiple printed circuit boards, and a single printed circuit board can contain one or multiple arrays **200**. For example, the array **200** of FIG. 2A is a multidimensional array **205**, and the array **200** of FIG. 2B is a one-dimensional array **210** or strip of light emitters **165** positioned generally in a straight line. The light emitting system **150** can comprise one or more such arrays **200**. For example, in one version, the array **200** can encompass the entirety or near entirety of the interior of the cuff **110**. In

another version, one or more arrays **200** can be positioned within the cuff **110** so that they each apply light to a particular region or muscle of interest on the body part within the cuff **110**. For example, in one particular version, one or more arrays **200** of light emitters **165** can be attached to the cuff **110** in independently controlled groups that correspond to target leg muscle groups such as quadriceps, hamstrings, and/or calves. Alternatively or additionally, other muscle groups or other tissue can be targeted. In one version, the one or more arrays **200** are individually integrated circuits mounted to flexible printed circuit boards that are attachable to the cuff **110**. In one version, one or more arrays **200** are connected in series to a main two connector power wire that is connected to the light emitting controller **175**.

[0051] The light emitting controller **175** controls the operation of the one or more arrays **200** of light emitters in a predetermined manner. The light emitting controller **175** supplies power to the one or more light emitters **165** so that a predetermined intensity and/or duration of light can be administered to a body part near or contacting the light emitter. The power and/or duration can be applied to a single light emitter **165** or to a plurality of light emitters **165**. When a plurality of light emitters **165** are to be powered by the light emitting controller **175**, they may be powered independently or as one or more arrays **200** where each array **200** includes a group of light emitters **165** that are to be powered collectively as a group at the same or similar intensity and for the same or similar duration. The intensity and/or the duration of the application of light for each light emitter **165** and/or for each array **200** can be input into the controller by a user communicating with the controller via a user interface. Alternatively, the intensity and/or duration can be automated. For example, in one version, the light emitting controller can be in communication with the pressure controller **125** or other monitor and can adjust the application of light in response to the pressurization of the cuff **110**. For example, by applying light to the body part only when the cuff **110** is pressurized, the application of light can be more efficient since it is during this pressurized condition that the light more optimally is transmitted to the muscle **187**. Alternatively, the light can be maintained even when the cuff **110** is in an unpressurized condition. Also alternatively or additionally, the intensity and/or duration can be adjusted in response to other signals or conditions as discussed below.

[0052] In one version, the intensity and/or duration of the application of light by a light emitter **165** and/or one or more arrays **200** of light emitters **165** can be selected or adjusted based on a factor associated with the body part to be treated. For example, one factor that can be used to adjust the intensity and/or duration of the light application is the thickness of the fatty layer **186** of the body part. Because the scattering of light is especially prevalent in fatty tissue, the thicker the layer of fat **186**, the higher the scattering of light and the greater the needed intensity and/or duration of light application to achieve a desired light application dosage. Accordingly, in one version, the thickness of the fatty layer **186** of the body part can be measured, such as by calipers or by an electronic monitor, and the intensity and/or duration of the light application can be set depending on the measurement. An operator of the light therapy system **100** can, for example, input the desired intensity and/or duration into the light emitting controller **175** after referring to a chart, table, or the like, that has empirical data relative to the fat layer condition, or the light emitting controller can be preprogrammed with the empirical data so that the operator can input the measurement information and the light emitting controller **175** can automatically adjust the intensity and/or duration of the light application. Alternatively, when using an electronic monitor, a signal related to the fat level condition can be supplied to the light emitting controller **175** and the adjustments can be made in response to the signal. Another factor that can be used to adjust the intensity and/or duration of the light application is skin color. Skin color can affect the transmissivity of the light application. Therefore, in one version of the invention, skin color can be assessed or monitored, and the light intensity and/or duration can be adjusted in accordance with the skin color in similar manner to that described above in connection with the fat condition. In yet another version, both fat layer thickness and skin color can be used to adjust the intensity and/or duration of the light intensity.

[0053] In one version, the light therapy system **100** may also include a light sensing system **300**, such as the one shown in FIGS. **3A** and **3B**. In the version of FIGS. **3A** and **3B**, the light sensing system comprises a photometer **305**, such as a reflectometer. The photometer **305** is made up of a photodetector **310** and a plurality of photometer light emitters **315**, such as infrared light emitting diodes. The photodetector **310** and the photometer light emitters **315** are secured to a substrate **320**. The photometer **305** comprises the photodetector **310** and a pattern of the photometer light emitters **315** that are spaced at logarithmic distances away from the photodetector **310**. For example, the photometer light emitters **315** may be infrared light emitting diodes that are placed at 3 mm, 6 mm, 12 mm, 24 mm, and/or 48 mm distances from the photodetector **310**. The photodetector **310** may include an optical filter on the sensor to detect only light within a narrow wavelength around the peak emitted wavelength of the infrared light emitting diodes **315**. In one, such as the one shown in FIGS. **3A** and **3B**, five photometer light emitters **315** and the photodetector **310** are mounted on a rigid substrate **320**, as shown, so that their relative spacing is maintained when the cuff **110** is inflated. Photometer light emitter power lines and sensor wires may be bundled together so the photometer **305** is in communication with the light emitting controller **175**. The light emitting controller **175** can control the emission of light from the photometer light emitters **315** and can receive an output signal from the photodetector **310**. The light emitting controller **175** can then process the data to determine qualities related to the transmission of light into the leg **115** or other body part. An optional pressure sensor **330** may also be provided as discussed below.

[0054] In use of the photometer **305** of FIGS. **3A** and **3B**, infrared photons from the photometer light emitters **315** are applied to the skin **185**. The photons are scattered, absorbed, or reemitted at the skin surface. The skin **185** has both light scattering and absorbing properties. The absorption is greater on darker skin with an abundance of melanin. Fat **186** may be approximated at mostly scattering of the infrared light. Muscle **187** tissue is primarily absorbing. The attenuation of light at the shortest distance between a first photometer light emitter **325** or group of photometer light emitters and the photodetector **310** is highly correlated with the skin light absorption since typical skin thicknesses on the torso and limbs is similar to 3 mm distance between the first photometer light emitter **325** and the photodetector **310**. The skin light absorption may be obtained from the empirical relationship:

skin absorption (cm.sup.-1)=f(photodetector voltage from first light emitter)

[0055] Light that penetrates the skin layer is scattered in all directions by the fat layer **186**. Some of this light exits the body back through the skin **185** and may be detected by the photodetector **310**. Light that passes through the fat layer **186** is absorbed by the muscle **187** where it can provide the desired therapeutic effect. Over thick fat layers, the amount of light remitted at the skin **185** will be higher than over thin fat layers **186** where light can travel less obstructed to the muscle tissue **187**. Each photometer light emitter is turned on and off in sequence so the output voltage from the photodetector **310** at different times corresponds to the diffuse reflectance at a discrete distance on the skin surface. The fat layer **186** thickness is inferred from the multiple measurements of the photometer light emitters **315** at varying distances from the photodetector **310**. The fat layer **186** thickness may be calculated from the empirical relationship:

fat thickness (mm)=g(photodetector voltage from all light emitters)

[0056] Another version of a photometer **305** of the light sensing system **300** of the light therapy system **100** is shown in FIG. **3C**. The photometer **305** of FIG. **3C** includes a circular geometric pattern **335** of photometer light emitters **315** with the photodetector **310** in the middle of the circular pattern surrounded by strings of one or more photometer light emitters **315**. Each string corresponds to a unique distance between the string and the photodetector **310** and is independently controllable by the light emitting controller **175**. The current through each string of photometer light emitters **315** is modulated to provide a response at the photodetector **310** that is approximately

consistent for a typical skin color and fat thickness. The photodetector voltage response may be near the middle of the range for an analog to digital converter to accurately measure the diffuse reflectance with a 10+bit resolution.

[0057] In one version, the light sensing system **300** can be used independently from the light application. For example, prior to the installation of the cuff **110** on the body part, the light sensing system **300** can be used to make measurements of the body part, such as the fat layer thickness and/or the light transmissivity. The measurements can be delivered to, input into, and used to set the light emitting controller **175**, as discussed above.

[0058] In one version, the light sensing system **300** can be integrated into the cuff **110** by being positioned within the cuff **110** and be directly connectable to the light emitting controller **175**. In this version, the photometer **305** can be used to make measurements of the body part prior to the application of light to the body part so the intensity and/or duration of the application can be adjusted. Additionally or alternatively, the photometer **305** can be used during the application of light. In one particular version, the photometer **305** can be used to monitor the transmissivity of light during the application, and the measurement can be interpreted by the light emitting controller **175** so that automated adjustments can be made to the intensity and/or duration of the light application. One or more photometers **305** can be positioned within the cuff **110** at a position in proximity to the one or more arrays **200** of light emitters **165** so that the intensity and/or duration of the light application for each array **200** can be separately adjusted in response to the measurement from the photometer associated with the region where the array **200** is located. The adjustments to the intensity and/or duration can be made by an operator receiving a signal indicative of the measurements or can be automatically made by light emitting controller **175** in response to the signal from the photometer **305**. In one version, the adjustments are made by the light emitting controller **175** in real time. For example, in one version, a photometer **305** may be positioned or mounted near the center of each of the one or more arrays **200** of light emitters **165**.

[0059] In one version, such as shown in FIG. **1** and in **4**, the light therapy system **100** may include a pressure cuff system **105** that operates as an intermittent pneumatic compression (IPC) device **400**. Intermittent pneumatic compression devices are used medically to help prevent clots in deep veins in the legs or other extremities. The intermittent pneumatic compression device **400** intermittently applies pneumatic pressure to the leg or other extremity or body part. Upon release of pressure, blood flows back through the veins and helps prevent blood clots. Intermittent pneumatic compression devices can also or alternatively be used by athletes to improve circulation and facilitate muscle recovery after exercise and/or for automated massage treatment.

[0060] In the version of FIG. **4**, the pressure cuff system **105** includes a cuff **110** having multiple chambers **405** that are each pneumatically separated from one another. When being used as an intermittent pneumatic compression device **400**, the pressure controller **125** can separately pressurize and depressurize each separate chamber **405** for a desired effect, such as to sequentially pressurize the chambers **405** in series to help encourage the flow of blood. In another version, the pressure controller **125** can pressurize each chamber **405** at the same time but with each chamber **405** being pressurized a different amount, as desired.

[0061] In FIG. **4**, there is further shown an exemplary light emitting system **150** having two or more light emitting members **155** that span multiple compartments **405** and a light sensing system **300** comprising a photometer **305** positioned between the light emitting members **155**. In an alternative arrangement, each light emitting member **155** can be associated with a separate compartment **405** and separate photometers **305** can be provided for multiple light emitting members **155**. The light emitting controller **175** is in electrical communication with each light emitting member **155** by, for example, respective wires **410**, **415** and is in communication with the one or more photometers **305**, such as by bundled wire **420** that can both power the photometer light emitters **315** and receive an output signal from the photodetector **310**. The light emitting controller **175** may include a processor and/or circuitry that controls the light emitting members

155 and may include input devices such as a touchscreen for receiving operator input. The light emitter power supply voltage and the air pump **130** may be connected to a mains power supply or a rechargeable battery for drawing electrical power, and provides the electrical power to the light emitters **165** as desired. The light emitting members **155** may be arranged in electrical series groups so that the sum of forward voltages of the group's light emitters **165** match the power supply voltage. The input device, such as the touch screen, may be utilized by a user to set a desired muscle infrared dose, for example.

[0062] The cuff **110** of the light therapy system **100** may optionally include a side opening **425**. The side opening **425** may be selectively openable to expose the interior of the cuff **110** and to facilitate attachment of the cuff **110** onto a leg or other extremity or other body part. A closure **430**, such as a zipper or the like, may be provided to close the opening **425** once the cuff **110** is secured around the body part.

[0063] FIG. 5 shows a particular cuff **110** and light emitting system **150**. The light emitting system **150** of FIG. 5 includes multiple strips of light emitters **165**, such as strips of light emitting diodes. In this version, each strip can operate as an array **200** or as a one-dimensional array **210** of light emitters **165**. Alternatively, multiple strips can be grouped together to form a larger array **200** or a multidimensional array. Alternatively, the entirety of the strips can collectively form a single array **200**. Each one-dimensional array **210** can contain from 2 to 100 light emitters **165**. In one version, there can be from 2 strips to about 50 strips, or from about 10 strips to about 40 strips, or from about 20 strips to about 30 strips, or about 26 strips. The total number of light emitters **165** can range from 2 to about 20,000, or from about 50 to about 10,000, or about 1000 to about 5000, and in one particular version there are about 4800 light emitters. The photometers **305** can optionally be mounted near the center of an array **200** of independently controllable light emitters **165**. The total number of photometers **305** can range from 0 to 20, or from about 1 to about 5, and in one particular version, there are four photometers **305**, one each for the right and left side of the calf and for the right and left side of the quadricep.

[0064] FIG. 6 shows an exploded view of a version of a pressure cuff system **105** of the light therapy system **100** with the cuff open and laid flat. The pressure cuff system **105** includes a cuff **110** that serves as an outer layer **600**. An intermediate layer **605** includes the one or more light emitting members **155** and the supporting structures **160** and the light sensing system **300**. Optionally, the intermediate layer **605** can further include a reflective liner **610**, such as a white liner. The reflective liner **610** can be attached to the cuff **110** and can be cut to be the same size and shape as the cuff **110** or can be different size and shape. The attachment of the reflective liner **610** to the cuff **110** may be in the form of zippers, hook and loop fasteners, or snap tape around the edges of the cuff **110** and/or each cuff chamber **405** or each cuff chamber that allow for convenient removal and servicing. The intermediate layer **605** may use zippers or other fasteners that mate with commercially available cuffs. Each side of the intermediate layer **605** may have two fasteners, one to attach to the cuff **110** and another to close the cuff **110** and intermediate layer **605** around a body part. The reflective or white liner **610** can improve the uniformity of the light dosage reaching the muscles. The light that is scattered back through the skin may be reflected back into the body by the reflective liner **610** to increase the chance of photons reaching the muscle tissue more efficiently. An inner layer **615** may also be provided. The inner layer **615** may be made of a material that is optically transparent to infrared light, such as polyvinyl chloride, polyurethane, epoxy, acrylic, silicone, and the like, and is attached to the intermediate layer **605** as a sheet with a fastener material or may be permanently attached thereto. The purpose of the optically transparent inner layer **615** is to insulate the user from the electrical components and provide a cleanable surface that can be disinfected and wiped down between uses. The permanently attached version of the inner layer **615** can be adhered to the intermediate layer **605** to create a durable product that provides protection for the light emitters **165** and photometers **305** from abrasion and the like. In this version, the inner layer **615** makes up the optional cover layer **195** discussed above. The light

emitting members **155** may be arranged in the electrical series groups so that the sum of forward voltages of the group's light emitters match the light emitter power supply voltage. The one or more series groups are connected to the main two connector power wire that is ultimately connected to the light emitting controller **175** that may be separated from the intermediate layer **605**. In one version, the light emitter members **155** may contain infrared light emitting diodes mounted to heat dissipating aluminum printed circuit boards that are attached to the white liner **610** using removable fastener or permanent attachment.

[0065] In one version, the intermediate layer **605** and optionally the inner layer **610** can be provided as an upgrade kit for an existing intermittent pneumatical compression device cuff assembly. Alternatively, the pressure cuff system **105** can include the cuff **110** and additional layers constructed together as a single unit.

[0066] In one version of the light therapy system **100**, a photometer **305** may be mounted near the middle of each muscle group array **200** so the dose delivered by the light emitters **165** on the array **200** may be controlled and/or customized in accordance with the signal received from the photometer **305**, as shown in FIG. **6**. Alternatively, multiple photometers **305** can be associated with each array **200**, such as by being distributed around the periphery or at respective ends of the array **200**. Alternatively, a single photometer **305** can be provided at a desired location.

[0067] FIG. **7** illustrates an example of a user interface **700**, such as a touch screen **705**, that may be used to allow a user to interact with the control system **120** and in particular the light emitting controller **175**. The user interface **700** shows parameters and status of infrared dosing for the light therapy system **100** applied to a leg **115**. The user interface **700**, for example, can be utilized by the user to specify the desired muscle IR dose (in joules per cm.sup.2) for each muscle group by using of a change tab or button. In real time, a display of the touch screen **705** user interface **700** may indicate one or more parameter values corresponding to fat thickness, skin light absorption, target muscle dose, progress toward the delivered muscle dose for each muscle group, and time remaining to complete the dosage. The touch screen **705** may further facilitate or allow or include a mechanism that will allow the user to reset the delivered dose counter and start and pause the ongoing dosing by way of other tabs or buttons **715**.

[0068] FIGS. **8A** and **8B** show different versions of pressure cuff systems **105** of the light therapy system **100**. FIG. **8A** shows the pressure cuff system **105** in the form of intermittent pneumatic compression shorts **800** that cover large muscle groups in upper legs and buttocks. In this version, the muscle group arrays **200** are applied in compression shorts **800** that cover the midsection of the body. When the middle seam **805** of the shorts **800** are opened and laid flat, the muscle group arrays **200** are applied in regions that correspond with the gluteus maximus, hamstrings, and quadriceps. FIG. **8B** shows a version of the pressure cuff system **105** integrated into or positionable on a massage chair **810** where the muscle group arrays **200** and light sensing systems **300** are integrated into installed on the chair **810** to treat muscle groups in arms, legs, neck, and/or back of a human body. In one version, the muscle group arrays **200** may be mounted in the massage chair so that the user receives the therapeutic light dose when the massage elements are creating maximum pressure on the targeted muscle groups. Commercially available massage chairs treat neck, shoulder, back, buttocks, arm, thigh, and calf muscles. The muscle group arrays **200** and light sensing systems **300** may be installed in a plurality of these locations as a kit where the array and sensors interface with the massage chair controls. Alternatively, the light therapy system **100** may be directly incorporated in the design of a massage chair. Alternatively, the light therapy system **100** may be an integral part of other structures, such as a massage table, or the like.

[0069] In operation, the light therapy system **100**, in one version, can employ a variety of programs that inflate the pressure cuff system **105** in a desired manner. Target cuff pressures are achieved either by inflating the cuff **110** and/or compartments **405** in the cuff **110** for a specific period of time or by monitoring the line pressures connected to the cuff **110**. When the maximum cuff pressure is achieved, a first solenoid valve or the like, such as a solenoid from one of the pressure

gates **145** connecting a pump, such as the electrical air pump **130** to the cuff **110** may close or the pump may stop. A second solenoid valve or the like, such as another solenoid from the pressure gate **130**, may vent the cuff **110** or compartment **405** to release the pressure according to a program schedule. The photometer **305** light emitting controller **175** may connect to the pressure controller **125** to monitor and control voltages corresponding to the state of the solenoid valves, the pressure sensor **140**, and the pump motor. Based on these control signals, a processor within the light emitting controller **175** determines a state when the cuff **110** or each compartment **405** is at a maximum pressure. The photometer **305** and the array **200** of light emitters **165** may operate when they are located underneath the cuff **110** that is at a desired and/or predetermined pressure state. When the solenoid vents the air in the cuff **110** or compartment **405**, both the photometer **305** and the arrays **200** may be disabled. A typical session with an intermittent pneumatic compression system may last approximately 15 minutes. The pump may go through multiple cycles that may last a few minutes each where the cuff **110** or compartments **405** are inflated to the certain pressure and the pressure is held for 5 to 30 seconds before venting the cuff **110** or compartment **405**. The light emitting controller **175** may, in one version, wait until a first predetermined pressure event occurs and make readings from the photometer **305** to characterize the light absorption and scattering of the skin and fat layers. These measurements may be used to calculate a dose of the IR light on the surface of the skin that will provide a uniform dose of the IR light to the muscles that will be absorbed in the muscle layer. The skin dose (DS) may have the form as shown below:

$$DS=DT*h(\text{skin absorption, fat thickness})$$

Where DT, dosage time, is the cumulative period when the light emitters are activated.

[0070] While the cuff **110** or compartment **405** is still in the pressurized condition, the array **200** of light emitters **165** may activate to begin the infrared light dose. The processor of the light emitting controller **175** may log the number of seconds for which the array **200** of light emitters are turned ON. The light emitters **165** may then be turned OFF when the corresponding cuff solenoid vents and turns back ON when the cycle reaches the predetermined pressure and/or duration. When the timer reaches the calculated exposure time for the target dose of the array **200**, the lights may turn OFF for the remainder of the program schedule.

[0071] In operation, each array **200** of light emitters **165** includes a number of light emitters **165** that emit light in a range of wavelengths spanning a central wavelength. The radiant flux (mW) of each light emitter is a measure of the emitted light in the range of wavelengths through an active area on the chip. The light emitter radiant power density is the radiant flux divided by the active area (mW/cm²). The light emitter radiant power density applies only to a point where the active area of the light emitter is in contact with the skin. For each array **200** and for each associated muscle group, a surface radiant power density is calculated. The surface radiant power density may be calculated as the number of light emitters **165** in the group multiplied by the radiant flux of an individual light emitters divided by the area treated by the group or array **200**. Beneath the skin and fat layers, the muscle radiant power density is the measure of how much of the light reaches the underlying muscle. The spacing of the light emitters on the array **200** of the muscle group light assembly can be important for proper therapeutic light dosing since too few lights may create dark regions between lights where the light does not propagate through to the muscle layer. Each array **200** has the light emitters spaced to deliver a nearly homogeneous power density to the target muscle. The ensemble of the light emitters may have sufficient radiant power to deliver the dose within the allotted time where the intermittent pneumatic compression program has the underlying cuff in the predetermined pressure state.

[0072] For areas of the body with thicker fat layers, less radiant power reaches the muscle group. The light reaching the muscle is also more diffuse. More powerful light emitters **165** may be spaced further apart but still achieve the desired homogeneity and density of light at the muscle layer. For example, a typical thickness of fat on a male calf (gastrocnemius muscle) is 4 mm+/-2

mm whereas the fat thickness on the inner thigh (adductor muscle) is 10 mm $\pm$ 4 mm. To deliver a homogeneous 3 J/cm² dose of the infrared radiation to both muscle groups in a 30 s exposure time, the inter light emitter spacing over the calf muscle group assembly may need to be 2 cm and use 200 mW radiant power light emitters and the light emitter spacing over the adductor may be 5 cm and use 1000 mW light emitters.

[0073] In another version, specific focusing and/or diffusing lenses may be placed over the light emitters **165** to help ensure uniform light density on the muscle surface. With this version, the array **200** of muscle group light emitters may be customized for different fat layer thickness without changing the light emitter spacing. The array **200** of muscle group light emitter assemblies could then be moved to target different muscle groups for a variety of intermittent pneumatic compression cuff assemblies, for example hand, foot, arm, legs, hips, thighs, and/or waist. In another version, a pressure sensor **330** is mounted on each photometer **305**. The array **200** may be turned ON when a pressure above a threshold value is sensed above the associated muscle group. The dosing may continue while the pressure is above the threshold value and the total exposure time is less than that needed to achieve the desired muscle dose. In another version, the light emitting controller **175** uses pulse width modulation (PWM) to control the dosage of light. The light emitting controller **175** may determine a duty cycle of the light emitters **165** associated with each muscle group so that the dosage of all muscle groups is completed at the same time.

[0074] Modeling of light transport through multiple tissue layers is a complex problem that can be addressed with statistical simulations that model the transport and fate of a large number of photons entering the body normal to the skin surface and interacting with the various tissue layers. The inputs of Monte Carlo simulations include optical properties of each tissue layer and their corresponding thicknesses. The output is a map of where photons are absorbed in tissue or re-emitted out of the skin. A summary of the modeled fate of IR photons for multiple simulations (changing both melanin content ranging from fair to dark skin color and fat thickness from 3 mm to 10 mm) is shown in FIG. 9A. In this simulation, the amount of light absorbed in the muscle layer ranges from 17% for fair skin and 3 mm fat layer to 3.2% for dark skin and 10 mm fat layer. To achieve the same dose of IR light to the underlying muscle, the later subject would need to expose themselves to more than 5 times the amount of light than the former subject. Inferring how much light reaches the muscle is helpful for delivering a consistent dose.

[0075] The model also generates a profile of how light is re-emitted from the skin surface as a function of distance from the light emitter light source, FIG. 9B. For points very close to the light emitting source ($x < 4$ mm, point A in FIG. 9B), the diffuse reflectance, $R_{\text{sub.d}}(A) = \text{detector signal} / \text{distance}_{\text{sup.2}}$, is closely related to how much light is not absorbed by the melanin in the epidermis layer. At source-detector distances on the order of the dermis thickness, the fat and muscle layers below the dermis have negligible effect on the $R_{\text{sub.d}}(A)$ signal. The skin factor, $F_{\text{sub.skin}}$ may be defined as $R_{\text{sub.d}}(4 \text{ mm})$ and represents a characteristic of the diffuse reflectance curve.

[0076] Light that penetrates the skin layer is scattered in all directions by the fat layer. Some of this light exits the body back through the skin and may be detected by the photodetector **310**. Light that passes directly through the fat layer is absorbed by the muscle where it can provide the desired effects of the light therapy. Over thick fat layers, the amount of light reemitted at the skin will be higher than over thin fat layers where light can travel with less obstruction from the photometer light emitters **315** to the light absorbing muscle tissue. The output voltage from the photodetector **310** at different times corresponds to the diffuse reflectance across a range of distances on the skin surface. The fat layer thickness is inferred from the multiple measurements light emitted from the skin **185** and varying distance from the source. Experiments have shown that further from the light emitter **315** source (> 5 mm), $R_{\text{sub.d}}(x)$ decays exponentially where the rate of decay is related to the thickness of the fat layer. The fat factor may be simply defined as the ratio $R_{\text{sub.d}}(10 \text{ mm}) / R_{\text{sub.d}}(20 \text{ mm})$ cancels out the component of the signal that is related to skin absorption.

[0077] For each model simulation, the skin and fat factors are tabulated and a collection of model runs with a range of different skin color and fat thickness layers creates a database for inferring the fraction of light absorbed by the muscle tissue, as shown by the following table of a sample database to infer infrared treatment time to deliver a constant dose of infrared light to muscle tissue based on variations in skin color and fat layer thickness.

TABLE-US-00001 Fair Skin- Fair Skin- Dark Skin- Dark Skin- 3 mm Fat 10 mm Fat 3 mm Fat 10 mm Fat Fitzgerald Skin 1 1 5 5 Color Type Fat Layer Thickness 3 10 3 10 (mm) Skin Factor, Rd 36.7 37.6 16.6 16.9 (4 mm) Fat Factor, 43.0 28.2 43.5 28.9 Rd(10 mm)/Rd (20 mm) Abs_muscle 17% 5% 11% 3% Treatment Time 10.0 35.3 14.7 51.2 (min)

[0078] Direct measurements of the reflectometer at distances of 4 mm, 10 mm, and 20 mm enable the calculation of both the skin and fat factors. In turn, the measured factors can be matched and interpolated with data from the above table to determine the proper treatment time that will deliver a constant dose of infrared light to muscle tissues for most users.

[0079] In other versions, the light therapy system **100** may use other parameters derived from the skin reflectometer's diffuse reflectance profile to infer adjustments to dosing treatment times. In addition, other data such as photometer sensor pressure **330** may be added to the database to improve the precision of infrared light dosing.

[0080] The power supply for the components and controllers may be integrated into a single enclosure and the light source is activated when the IPC pump has reached the predetermined pressure in the cuff **110**. In this way, the light is only emitted when the optical path from the light source module to the muscle is shortest. The controller can be set with a prescribed total infrared or other radiation dosage (J or J/cm²) for each muscle group so that the muscle group light source is turned OFF when the desired dosage has been achieved.

[0081] In one version of the invention, the light sensing system **300** can be used with a light therapy system other than one that uses a pressure cuff system. Light therapy of all types can benefit from a measurement of the thickness of a fatty layer and the adjustment of the light dosage in relation to the measurement. For example, the light sensing system **300** can be used to determine optimal light dosage on the skin to deliver the desired light dosage to the muscle tissue in any other mode of infrared light therapy. Other light therapy systems include light bulbs (Wolezek) and LED panels (i.e. Joovv, PlatimunLED Therapy Lights) where a subject stands in front of the light for a predetermined period of time. Other methods can also include tanning bed style systems (NovoThor). In addition, hand-held LED and laser IR emitters (Infarex, TOPlight) can be held on the skin surface to achieve therapeutic effects. Optical properties of skin color and fat thickness from clinical test subjects may be approximated and used as input to the multi layered Monte Carlo model. The model will determine what fraction of IR photons incident to the subject's skin are absorbed in the muscle tissue. Using the relationships between diffuse reflectance profile from the IR photometer output and the faction of light absorbed in the muscle layer (Abs muscle) from Table 1, an adjustment to the Treatment Time or light intensity can be calculated that will deliver a dose of IR light to the muscle tissue.

[0082] The control system **120** including the pressure controller **125** and/or the light emitting controller **175** can be any device capable of receiving input, performing calculations, performing calculations based on the input, producing an output signal, and/or producing an output signal as a result of the calculations. The controllers **125**, **175** may be part of the same controller or may be separate controllers that are capable of communication with one another. The controllers **125**, **175** may be in the form of a central processor that is capable of interacting with a user via a keyboard, a graphical user interface, wireless communication, voice command, or any other manner. For example, the controllers **125**, **175** may be a personal computer, a laptop, a handheld device, a server, a network of servers, a cloud network, or the like. The operator may interact with the controller **125** and/or controller **175** before, during, or after the light therapy procedure. The controllers **125**, **175** can include various modules that allow it to perform calculations, algorithms,

routines, and/or subroutines to process information and/or make determinations. The controllers 125, 175 may further include other optional modules, such as artificial intelligence and/or machine learning modules that use algorithms to parse data, learn from the data, and then to make determinations and/or predictions based on what was learned.

[0083] Although the present invention has been described in considerable detail with regard to certain preferred versions thereof, other versions are possible, and alterations, permutations and equivalents of the version shown will become apparent to those skilled in the art upon a reading of the specification and study of the drawings. For example, the cooperating components may be reversed or provided in additional or fewer number. Also, the various features of the versions herein can be combined in various ways to provide additional versions of the present invention. Furthermore, certain terminology has been used for the purposes of descriptive clarity, and not to limit the present invention. Throughout this specification and any claims appended hereto, unless the context makes it clear otherwise, the term “comprise” and its variations such as “comprises” and “comprising” should be understood to imply the inclusion of a stated element, limitation, or step but not the exclusion of any other elements, limitations, or steps. Therefore, any appended claims should not be limited to the description of the preferred versions contained herein and should include all such alterations, permutations, and equivalents as fall within the true spirit and scope of the present invention.

Claims

1-20. (canceled)

21: A light therapy system, comprising: a pressurizable cuff positionable on or near a body part of a user; a support structure coupled to a surface of the pressurizable cuff; a light emitter comprising (i) a bottom surface facing the support structure and (ii) a light emitting surface spaced apart from the support structure, the light emitting surface configured to emit light towards the user's body part and to be pressed against a skin of the user's body part when the cuff is pressurized; and a plastic non-adhesive cover layer disposed over the light emitter, the cover layer configured to aid in preventing direct contact between the light emitter and the skin of the user's body part and comprising (i) an outer surface that, at least in part, is configured to face the skin of the user's body part and (ii) an opposite inner surface, wherein at least a portion of a surface of the light emitter different than the bottom surface is free from direct contact with the cover layer's inner surface.

22: The light therapy system of claim 21, wherein the pressurizable cuff is positionable on a leg of the user.

23: The light therapy system of claim 21, wherein the pressurizable cuff comprises a flexible material.

24: The light therapy system of claim 21, wherein the pressurizable cuff comprises a side opening.

25: The light therapy system of claim 21, wherein the pressurizable cuff comprises a zipper.

26: The light therapy system of claim 21, wherein the support structure comprises a circuit board.

27: The light therapy system of claim 21, wherein the light emitter is a light emitting diode.

28: The light therapy system of claim 21, wherein the light emitting surface is configured to emit infrared light towards the user's body part.

29: The light therapy system of claim 21, wherein the cover layer further comprises a light transmitting material.

30: The light therapy system of claim 21 further comprising a controller for controlling a pressurization of the pressurizable cuff.

31: The light therapy system of claim 30, wherein the controller is configured to selectively pressurize and depressurize the pressurizable cuff.

32: The light therapy system of claim 30, wherein the pressurizable cuff comprises a plurality of chambers.

- 33:** The light therapy system of claim 32, wherein the controller is configured to separately pressurize and depressurize each chamber of the pressurizable cuff.
- 34:** The light therapy system of claim 32, wherein the controller is configured to sequentially pressurize each chamber of the pressurizable cuff.
- 35:** The light therapy system of claim 32, wherein the controller is configured to pressurize each chamber of the pressurizable cuff at the same time.
- 36:** The light therapy system of claim 21 further comprising a controller for controlling the emission of light from the light emitter.
- 37:** The light therapy system of claim 21 further comprising at least one array of light emitters, each array comprising a plurality of the light emitters positioned in a straight line.
- 38:** The light therapy system of claim 21 further comprising an input device for receiving user input.
- 39:** The light therapy system of claim 38, wherein the input device comprises a screen.
- 40:** The light therapy system of claim 21 further comprising a rechargeable battery.
- 41:** A light therapy system, comprising: a pressurizable cuff positionable on or near a body part of a user; a circuit board coupled to a surface of the pressurizable cuff; a light emitter comprising a light emitting surface, the light emitting surface configured to emit light towards the user's body part and to be pressed against a skin of the user's body part when the cuff is pressurized; a spacer disposed between the circuit board and the light emitter, such that (i) at least a first portion of the spacer is directly attached to at least a portion of a top surface of the circuit board and (ii) at least a second portion of the spacer is directly attached to the light emitter; and a cover layer in direct contact with the light emitting surface of the light emitter and extending over a section of the top surface of the circuit board, wherein at least a portion of the section of the top surface of the circuit board over which the cover layer extends is free from direct contact with the cover layer and is free from direct contact with the spacer, and wherein the cover layer is configured to prevent direct contact between the light emitter and the skin of the user's body part.
- 42:** The light therapy system of claim 41, wherein the pressurizable cuff is positionable on a leg of the user.
- 43:** The light therapy system of claim 41, wherein the pressurizable cuff comprises a flexible material.
- 44:** The light therapy system of claim 41, wherein the pressurizable cuff comprises a side opening.
- 45:** The light therapy system of claim 41, wherein the pressurizable cuff comprises a zipper.
- 46:** The light therapy system of claim 41, wherein the light emitter is a light emitting diode.
- 47:** The light therapy system of claim 41, wherein the light emitting surface is configured to emit infrared light towards the user's body part.
- 48:** The light therapy system of claim 41, wherein the cover layer comprises a light transmitting material.
- 49:** The light therapy system of claim 41 further comprising a controller for controlling a pressurization of the pressurizable cuff.
- 50:** The light therapy system of claim 49, wherein the controller is configured to selectively pressurize and depressurize the pressurizable cuff.
- 51:** The light therapy system of claim 49, wherein the pressurizable cuff comprises a plurality of chambers.
- 52:** The light therapy system of claim 51, wherein the controller is configured to separately pressurize and depressurize each chamber of the pressurizable cuff.
- 53:** The light therapy system of claim 51, wherein the controller is configured to sequentially pressurize each chamber of the pressurizable cuff.
- 54:** The light therapy system of claim 51, wherein the controller is configured to pressurize each chamber of the pressurizable cuff at the same time.
- 55:** The light therapy system of claim 41 further comprising a controller for controlling the

emission of light from the light emitter.

56: The light therapy system of claim 41 further comprising at least one array of light emitters, each array comprising a plurality of the light emitters positioned in a straight line.

57: The light therapy system of claim 41 further comprising an input device for receiving user input.

58: The light therapy system of claim 57, wherein the input device comprises a screen.

59: The light therapy system of claim 41 further comprising a rechargeable battery.
